

Prelude and Fugue

1. Fugue

Christopher Arehart

$\text{♩} = 110$

Voice-Leading Flags (R# on notes = multiple rules)

3. [Indigo] Parallel unisons

2. [Vermilion Orange] Parallel octaves

1. [Crimson Red] Parallel P5ths

9. [Violet] Large leap

8. [Safety Orange] Melodic tritone

6. [Azure Blue] Voice crossing

10. [Golden Yellow] Leading tone unresolved

Soprano Recorder

Alto Recorder

Tenor Recorder

Bass Recorder

6

S. Rec.

A. Rec.

T. Rec.

B. Rec.

R2 R2

R10 R2 R2

R10

R10

R10

R6,9 R6 R6

R2 R2

R6 R6

6 4 7 7 6 7 4

R2 R2

7 5 6 4 7 5 9 9

R6

10

S. Rec.

A. Rec.

T. Rec.

B. Rec.

R1 R1 R10 R1 R1 R2 R2 R2 R2 R2 R2 R2 R10 R9

2 2 2 4 4 4 2 2 6 #6 #3 2 7

R1 R1 R6

14

S. Rec.

A. Rec.

T. Rec.

B. Rec.

R1,10 R1,2 R2 R6,10 R1 R1 R2,6 R2 R6 R6 R6,9R6 R6 R10

4 #3 9 9 7 4 3 4 3 6 4 2 #3 4 3 9 7 3 #7 7 #7

R9 R6 R6,8 R6

18

S. Rec.

A. Rec.

T. Rec.

B. Rec.

R2 R2 R1 R1 R6 R1 R1,10

R6 R10 R2 R2,6 R10

R6

R6 R2 R2 R1 R1 R2 R1,2 R1

#2 4 #9 7 9 5 4 6 5 9 4 3 9 7 3 4 #3 #2 #3

26

S. Rec.

A. Rec.

T. Rec.

B. Rec.

R6 R10 R2 R2 R9 R6

R6

R6

4 9 4 6 6 #6 #4 6 7 7 7 7 #7 7 6 7 6 7 6 R2 R2 R6,9 R6 R6

31

S. Rec.

A. Rec.

T. Rec.

B. Rec.

R9 R2 R2 R10

R2 R2 R6,10R1 R1

R10 R6

4 7 2 9 2 6 6 7 6 45 6

R6,9 R1 R1 4 4 5

35

S. Rec.

A. Rec.

T. Rec.

B. Rec.

R6

R6

R6,10

R6

7 7
R6 5

4 3
R6

3 7 9

9

4 3 9

R10

R6

R6

R6

Detailed description: This is a musical score for four voices: Soprano (S. Rec.), Alto (A. Rec.), Tenor (T. Rec.), and Bass (B. Rec.). The score is in G major (one sharp) and 4/4 time. Measure 35 is marked with a '35' above the Soprano staff. The Soprano part consists of half notes: G4, A4, B4, and G4. The Alto part starts with a quarter note G4, followed by eighth notes A4, B4, A4, and G4, then a half note G4. In measure 36, the Alto has a half note G4, a quarter note A4, a quarter note B4, and a half note G4. In measure 37, the Alto has a half note G4, a quarter note A4, a quarter note B4, and a half note G4. In measure 38, the Alto has a half note G4. The Tenor part starts with a quarter note G3, followed by eighth notes A3, B3, A3, and G3, then a half note G3. In measure 36, the Tenor has a half note G3, a quarter note A3, a quarter note B3, and a half note G3. In measure 37, the Tenor has a half note G3, a quarter note A3, a quarter note B3, and a half note G3. In measure 38, the Tenor has a half note G3. The Bass part starts with a quarter note G2, followed by eighth notes A2, B2, A2, and G2, then a half note G2. In measure 36, the Bass has a half note G2, a quarter note A2, a quarter note B2, and a half note G2. In measure 37, the Bass has a half note G2, a quarter note A2, a quarter note B2, and a half note G2. In measure 38, the Bass has a half note G2. The score includes various annotations: 'R6' (Root 6th) is written below the Soprano staff in measures 35, 36, 37, and 38. 'R6,10' (Root 6th and 10th) is written below the Alto staff in measure 37. 'R10' (Root 10th) is written below the Alto staff in measure 38. 'R6' (Root 6th) is written below the Tenor staff in measures 35, 36, 37, and 38. 'R6' (Root 6th) is written below the Bass staff in measures 35, 36, 37, and 38. The score also includes various accidentals and articulation marks, such as a natural sign under the Tenor's G3 in measure 35 and a fermata over the Alto's G4 in measure 38.

Voice Leading Analysis

Baroque Counterpoint Violations Report

Score:

Prelude and Fugue for Recorders in G - Full score - 01 Fugue.musicxml

Summary

Total Violations: 75

Rules Violated: 7

Color Legend

 Voice crossing (26)	 Large leap (8)
 Leading tone unresolved (21)	 Melodic tritone (1)
 Parallel octaves (10)	 Parallel unisons (1)
 Parallel P5ths (8)	

Detailed Violations

Voice crossing (26)

A lower voice temporarily moves above a higher voice, creating confusion in register.

m.9	beat 1.00	Alto Recorder & Tenor Recorder	Alto Recorder crosses Tenor Recorder (G vs C)
m.9	beat 2.00	Alto Recorder & Tenor Recorder	Alto Recorder crosses Tenor Recorder (D vs A)
m.9	beat 4.00	Alto Recorder & Bass Recorder	Alto Recorder crosses Bass Recorder (B vs C)
m.12	beat 2.50	Tenor Recorder & Bass Recorder	Tenor Recorder crosses Bass Recorder (A vs F#)
m.13	beat 3.00	Alto Recorder & Bass Recorder	Alto Recorder crosses Bass Recorder (G vs A)
m.15	beat 2.50	Alto Recorder & Tenor Recorder	Alto Recorder crosses Tenor Recorder (F# vs G)
m.16	beat 1.00	Tenor Recorder & Bass Recorder	Tenor Recorder crosses Bass Recorder (A vs C#)
m.16	beat 1.50	Alto Recorder & Bass Recorder	Alto Recorder crosses Bass Recorder (A vs G)
m.16	beat 1.50	Tenor Recorder & Bass Recorder	Tenor Recorder crosses Bass Recorder (B vs G)
m.16	beat 2.00	Alto Recorder & Tenor Recorder	Alto Recorder crosses Tenor Recorder (E vs C#)
m.16	beat 4.00	Alto Recorder & Bass Recorder	Alto Recorder crosses Bass Recorder (C# vs E)
m.18	beat 1.00	Alto Recorder & Tenor Recorder	Alto Recorder crosses Tenor Recorder (B vs A)
m.18	beat 1.00	Tenor Recorder & Bass Recorder	Tenor Recorder crosses Bass Recorder (A vs C#)
m.21	beat 1.00	Soprano Recorder & Alto Recorder	Soprano Recorder crosses Alto Recorder (D vs A)
m.22	beat 2.00	Alto Recorder & Tenor Recorder	Alto Recorder crosses Tenor Recorder (C# vs A)
m.23	beat 2.00	Alto Recorder & Tenor Recorder	Alto Recorder crosses Tenor Recorder (D vs E)
m.27	beat 3.00	Alto Recorder & Tenor Recorder	Alto Recorder crosses Tenor Recorder (F# vs D)
m.29	beat 4.00	Tenor Recorder & Bass Recorder	Tenor Recorder crosses Bass Recorder (A vs D)
m.30	beat 1.00	Tenor Recorder & Bass Recorder	Tenor Recorder crosses Bass Recorder (D vs G)
m.30	beat 2.00	Alto Recorder & Bass Recorder	Alto Recorder crosses Bass Recorder (D vs B)
m.32	beat 1.00	Alto Recorder & Bass Recorder	Alto Recorder crosses Bass Recorder (F# vs C)
m.32	beat 1.00	Tenor Recorder & Bass Recorder	Tenor Recorder crosses Bass Recorder (E vs C)
m.35	beat 1.00	Tenor Recorder & Bass Recorder	Tenor Recorder crosses Bass Recorder (G vs E)
m.36	beat 1.00	Tenor Recorder & Bass Recorder	Tenor Recorder crosses Bass Recorder (F# vs A)
m.37	beat 1.00	Tenor Recorder & Bass Recorder	Tenor Recorder crosses Bass Recorder (F# vs D)
m.38	beat 1.00	Alto Recorder & Tenor Recorder	Alto Recorder crosses Tenor Recorder (B vs D)

Leading tone unresolved (21)

The leading tone (7th scale degree) should resolve upward to the tonic.

m.3	beat 4.00	Alto Recorder	LT F# !' E (expected tonic G)
m.6	beat 1.50	Alto Recorder	LT F# !' E (expected tonic G)
m.7	beat 1.00	Alto Recorder	LT F# !' E (expected tonic G)
m.7	beat 2.00	Soprano Recorder	LT F# !' A (expected tonic G)
m.8	beat 2.50	Alto Recorder	LT F# !' E (expected tonic G)
m.10	beat 1.00	Tenor Recorder	LT F# !' E (expected tonic G)
m.11	beat 4.00	Soprano Recorder	LT F# !' E (expected tonic G)
m.13	beat 3.50	Tenor Recorder	LT F# !' A (expected tonic G)
m.15	beat 2.50	Alto Recorder	LT F# !' E (expected tonic G)
m.15	beat 3.00	Soprano Recorder	LT F# !' E (expected tonic G)
m.17	beat 2.50	Tenor Recorder	LT F# !' E (expected tonic G)
m.18	beat 3.00	Alto Recorder	LT F# !' B (expected tonic G)
m.21	beat 2.00	Soprano Recorder	LT F# !' D (expected tonic G)
m.21	beat 2.00	Alto Recorder	LT F# !' C# (expected tonic G)
m.23	beat 1.00	Soprano Recorder	LT F# !' D (expected tonic G)
m.28	beat 1.00	Alto Recorder	LT F# !' E (expected tonic G)
m.31	beat 2.50	Soprano Recorder	LT F# !' E (expected tonic G)
m.31	beat 3.00	Tenor Recorder	LT F# !' E (expected tonic G)
m.32	beat 1.00	Alto Recorder	LT F# !' A (expected tonic G)
m.37	beat 1.00	Tenor Recorder	LT F# !' E (expected tonic G)
m.37	beat 2.00	Alto Recorder	LT F# !' E (expected tonic G)

Parallel octaves (10)

Consecutive perfect octaves eliminate the distinction between voices.

m.6	beat 2.00	Soprano Recorder & Tenor Recorder	P8 !' P8 (A-A !' G-G)
m.7	beat 3.00	Soprano Recorder & Bass Recorder	P8 !' P8 (A-A !' G-G)
m.11	beat 4.00	Alto Recorder & Tenor Recorder	P8 !' P8 (C-C !' D-D)
m.13	beat 1.50	Soprano Recorder & Tenor Recorder	P8 !' P8 (C#-C# !' D-D)
m.13	beat 2.00	Soprano Recorder & Tenor Recorder	P8 !' P8 (D-D !' E-E)
m.16	beat 3.00	Soprano Recorder & Alto Recorder	P8 !' P8 (E-E !' D-D)
m.19	beat 1.00	Soprano Recorder & Bass Recorder	P8 !' P8 (C#-C# !' D-D)
m.21	beat 1.00	Alto Recorder & Bass Recorder	P8 !' P8 (G-G !' A-A)
m.29	beat 1.00	Alto Recorder & Bass Recorder	P8 !' P8 (G-G !' A-A)
m.31	beat 2.00	Soprano Recorder & Alto Recorder	P8 !' P8 (A-A !' G-G)

Parallel P5ths (8)

Consecutive perfect fifths in two voices create parallel motion that undermines voice independence.

m.4	beat 2.00	Soprano Recorder & Alto Recorder	P5 !' P5 (B-E !' A-D)
m.11	beat 3.00	Soprano Recorder & Tenor Recorder	P5 !' P5 (A-D !' G-C)
m.12	beat 2.00	Soprano Recorder & Bass Recorder	P5 !' P5 (E-A !' D-G)
m.16	beat 1.00	Soprano Recorder & Alto Recorder	P5 !' P5 (F#-B !' E-A)
m.19	beat 3.00	Soprano Recorder & Bass Recorder	P5 !' P5 (E-A !' D-G)
m.21	beat 2.00	Soprano Recorder & Bass Recorder	P5 !' P5 (E-A !' F#-B)
m.24	beat 4.00	Tenor Recorder & Bass Recorder	P5 !' P5 (C#-F# !' D-G)
m.32	beat 2.00	Alto Recorder & Bass Recorder	P5 !' P5 (A-D !' B-E)

Large leap (8)

A leap of 10+ semitones should be filled in by stepwise motion in the opposite direction.

m.9	beat 1.00	Alto Recorder	leap of 11 semitones: F# !' G
m.13	beat 4.50	Tenor Recorder	leap of 12 semitones: A !' A
m.15	beat 2.00	Bass Recorder	leap of 10 semitones: E !' D
m.16	beat 1.00	Tenor Recorder	leap of 12 semitones: A !' A
m.29	beat 4.00	Bass Recorder	leap of 12 semitones: D !' D

m.30 beat 1.00 | Alto Recorder | leap of 14 semitones: A !' G
m.31 beat 1.00 | Soprano Recorder | leap of 11 semitones: C# !' D
m.32 beat 1.00 | Bass Recorder | leap of 14 semitones: D !' C

Melodic tritone (1)

A tritone in melodic motion is dissonant and should be avoided in voice leading.

m.16 beat 1.50 | Bass Recorder | tritone leap: C# !' G

Parallel unisons (1)

Two voices in unison reduce the texture and blur voice independence.

m.25 beat 1.00 | Alto Recorder & Bass Recorder | unison !' unison on A

Rules Reference

Parallel perfect intervals

Two or more voices moving in parallel into a perfect 5th or octave undermines voice independence.

Parallel unison

Similar motion leading to unison eliminates register distinction.

Direct intervals (outer voices)

Similar motion into a perfect interval in the outer voices, especially with soprano leap.

Voice crossing

Outer voices exchange registers, creating confusion in the texture.

Melodic intervals

Augmented 2nds and tritones are awkward in a single voice.

Large leaps

Leaps "e10 semitones should be filled in by stepwise opposite motion.

Leading tone resolution

The 7th scale degree should resolve up to the tonic.

Voice-Leading Rule Review and Correction Guide

Source analyzed:

Prelude and Fugue for Recorders in G - Full score - 01 Fugue.musicxml

Summary from current report:

- Total flags: 205
- Leading tone unresolved: 33
- Voice crossing: 20
- Parallel unisons: 107
- Parallel octaves: 13
- Parallel P5ths: 17
- Large leap: 11
- Melodic tritone: 2
- Direct 5ths: 2

Important context:

Many flags mention Viola. If you intend to work with the reduced 4-recorder version only, re-run analysis after removing Viola so counts and examples match the current score exactly.

1) Leading Tone Unresolved (33)

Why this rule is flagged:

- In tonal counterpoint, the leading tone (scale degree 7) has strong tendency to resolve up by semitone to tonic.
- In G major, that means F# is expected to move to G in accented or cadential contexts.
- The analyzer flags cases where F# holds, descends, or leaps elsewhere instead of resolving to G.

Representative report examples:

- m.4 beat 1.00 (Alto): F# -> E
- m.7 beat 2.50 (Soprano): F# -> A
- m.21 beat 4.00 (Alto): F# -> C#

Why these may be musical anyway:

- In fugue texture, passing, neighbor, or sequence-driven lines may defer or avoid immediate tendency-tone resolution.
- Inner voices can sometimes move to chordal tones other than tonic if harmony is clear and voice leading remains smooth.

Correction options:

- Preferred: move F# to G in the next note in the same voice, especially at cadences and downbeats.
- If harmonic function requires otherwise, delay but eventually complete the tendency resolution by step.
- Revoice surrounding parts so F# becomes non-structural (passing tone) with weaker accent.

2) Voice Crossing (20)

Why this rule is flagged:

- A lower voice moves above a higher voice (or vice versa), blurring registral hierarchy.
- The revised detector already ignores unison landings and direct pitch swaps, so these are more likely true crossings.

Representative report examples:

- m.9 beat 1.00: Soprano crosses Alto; Alto crosses Tenor
- m.16 beat 2.00: Alto crosses Tenor
- m.38 beat 1.00: Alto crosses Tenor

Why these may be musical anyway:

- Imitative writing can tolerate brief crossings when subjects overlap.
- Some crossings are texturally intentional for color or registral exchange.

Correction options:

- Keep each line within its normal register envelope (S > A > T > B).
- When two voices converge, route one by contrary step while the other holds or moves less.
- Re-octavate one note in the inner voice to restore vertical order without changing harmony.

3) Parallel Unisons (107)

Why this rule is flagged:

- Two voices move together in repeated unisons over successive attacks (unison -> unison).
- This merges lines and reduces contrapuntal independence.

Representative report examples:

- m.6 beat 2.00: Soprano & Tenor unison -> unison
- Many sustained chains in Bass & Viola through middle sections

Why these may be musical anyway:

- If one line is a reinforcement doubling (especially in arrangements), repeated unisons can be deliberate.
- With Viola removed, many of these may disappear in the practical version.

Correction options:

- Convert one member of the pair to thirds, sixths, or oblique motion.
- Keep occasional unison arrivals but avoid repeated consecutive unison motion.
- Offset one voice rhythmically by suspending or anticipating its next pitch.

4) Parallel Octaves (13)

Why this rule is flagged:

- Two voices form a perfect octave on one attack and another perfect octave on the next while moving in similar motion.
- This weakens perceived independence similarly to parallel unisons.

Representative report examples:

- m.7 beat 3.00: Soprano & Bass (and Soprano & Viola)
- m.21 beat 1.00: Alto & Bass (and Alto & Viola)

Why these may be musical anyway:

- In thicker ensemble textures, occasional octave reinforcement can be stylistically tolerated.
- But repeated structural octaves are generally treated as an error in strict species-style counterpoint.

Correction options:

- Alter one voice by step to arrive at 10th/6th instead of 8ve.
- Use contrary motion into the second sonority.
- Reassign one voice note to neighboring chord tone.

5) Parallel Perfect Fifths (17)

Why this rule is flagged:

- Consecutive perfect fifths in similar motion between two parts.
- In strict common-practice pedagogy, this is one of the primary independence violations.

Representative report examples:

- m.11 beat 3.00: Soprano & Tenor
- m.21 beat 2.00: Soprano & Bass (and with Viola)
- m.32 beat 2.00: Alto & Bass

Why these may be musical anyway:

- In sequential passages, temporary fifth chains may arise from pattern transposition.
- Still usually worth reducing if you want stricter pedagogical compliance.

Correction options:

- Change one target note by step to produce a 6th/3rd at the second sonority.
- Introduce contrary or oblique motion in one voice between the two chords.
- Break sequence literalism for one event to avoid the repeated perfect interval.

6) Large Leap (11)

Why this rule is flagged:

- Melodic leaps of 10+ semitones (minor seventh or larger), including octave-plus motions.
- Large skips are harder to sing and often demand compensating stepwise recovery.

Representative report examples:

- m.30 beat 1.00 (Alto): leap of 14 semitones
- m.32 beat 1.00 (Bass): leap of 14 semitones

Why these may be musical anyway:

- Instrumental fugues can support larger leaps than vocal writing.
- Octave transfers can be expressive if balanced by contour and phrase design.

Correction options:

- Replace large leap with two smaller intervals via passing notes.
 - If keeping the leap, follow with stepwise motion opposite the leap direction.
 - Redistribute octave transfer to another voice where tessitura is more natural.
- 7) Melodic Tritone (2)

Why this rule is flagged:

- A single voice leaps by tritone (augmented 4th/diminished 5th), often considered harsh in strict melodic writing.

Representative report examples:

- m.16 beat 1.50: Bass (and Viola in full-score analysis) C# -> G

Why these may be musical anyway:

- In harmonic minor/sequence contexts, tritone outlines can occur idiomatically in instrumental lines.

Correction options:

- Fill the span with passing motion instead of direct leap.
- Redirect one endpoint to nearest chord tone producing a perfect 4th/5th or major/minor 3rd.
- Keep harmonic target but adjust rhythm to approach/leave by step.

8) Direct (Hidden) Fifths (2)

Why this rule is flagged:

- Outer voices move in similar motion into a perfect fifth, with upper voice leaping.
- This can mimic the effect of parallels even without literal consecutive perfect intervals.

Representative report examples:

- m.16 beat 2.00: Alto & Bass (outer-voice interpretation in current indexing)
- m.32 beat 1.50: Alto & Bass

Why these may be musical anyway:

- In fast contrapuntal textures, some hidden perfects are acceptable if weakly accented and linearly justified.

Correction options:

- Change upper voice approach to stepwise motion.
- Approach target by contrary motion in one outer voice.
- Shift one note to a non-perfect interval at arrival.

Recommended workflow to fix efficiently

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1. Prioritize structural errors first: Parallel P5ths, Parallel Octaves, Direct 5ths.
 2. Next clean tendency-tone handling at cadences: Leading tone unresolved.
 3. Then smooth local line shape: Large leaps and Melodic tritones.
 4. Treat Parallel unisons and Voice crossings with musical discretion in imitative passages.
- 5-minute triage pass:
- Focus measures: 7, 9, 16, 21, 32.
 - In each, first remove repeated perfect intervals in outer parts.
 - Then ensure any accented F# resolves to G unless intentionally sequenced.
- End of review.